Appendix B

Background on basic logic

Logical implication

Basic logic begins with statements of logical implication of the form "A implies B". We write it $A \Longrightarrow B$. The converse is $B \Longrightarrow A$ and is logically independent of the original statement, which means they could both be true, both false, or one false and one true.

If a statement $A \Longrightarrow B$ and its converse $B \Longrightarrow A$ are both true then A and B are logically equivalent, and we say "A iff B" (short for "if and only if") or use the double-headed arrow $A \Longleftrightarrow B$.

Another logical statement related to $A \Longrightarrow B$ is the *contrapositive*, which is the statement "not $B \Longrightarrow$ not A". This is logically equivalent to $A \Longrightarrow B$. (Note the reversed direction.)

Quantifiers

The expressions "for all" and "there exists" are called quantifiers, and we notate them by $\forall$ and $\exists$. It is important to understand how to negate them. The negation of a statement is the statement of its untruth. We can always negate a statement A by saying "not A", but sometimes this can be unraveled to something more basic. Here is a table for negating quantifiers involving statements A and B:

original:	$\forall\, x$	A
negation:	$\exists\, x$ such that	not A
original:	$\exists\, x$ such that	B
negation:	$\forall\, x$	not B

When we use quantifiers in a formal logical statement it might seem that we need some commas, but we often omit the punctuation because it gets in the way without really clarifying anything. Here's an example:

$$\forall\, b \in B \quad \exists\, a \in A \ \text{ such that } \ f(a) = b.$$